Tianle (Terrence) Liang

ttliang@engineering.upenn.edu // (317)-518-6395 // LinkedIn // Portfolio

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA *May 2030*
Candidate for Bachelor of Science in Engineering (BSE)
Major: Artificial Intelligence, concentration in robotics

Zionsville Community High School, Zionsville, Indiana *Jun 2026*
SAT: 1550; Ranked 5/575; National Merit Scholar; Coolidge Senator

EXPERIENCE

Systems Optimization Laboratory (SOL) | *Energy Optimization Researcher*, Stanford, CA *July 2025 - Present*

- Built nonlinear AMPL model allocating resilience investments in 18 ERCOT plant clusters (7 fuel sources, 10 grid cells), against $43.1B in modeled 2025-2050 losses
- Optimized using SNOPT (resulting in $5B budget cap, 2.1x ROI); 300-trial Monte Carlo held >1x ROI for all trials
- Mentored by Stanford prof. Michael Saunders and shipped public simulator (in process of addition to SOL webpage)

Rayfield Systems (Rayfield Holdings) | *Technical Intern*, Stanford, CA *Jun 2025 - July 2025*

- Led 8-intern team building ML pipeline (R^2 of 0.94, 85% accuracy) for wind generation and anomaly forecasting (up to 336-hour horizon) on a farm's 43K hourly records (2017-2021)
- Presented directly to CEO Pranav Kulkarni (Stanford SAIL researcher); part of Harvard Ventures TECH program

NextGen Research Institute | *Chief Technology Officer*, Indianapolis, IN *Nov 2024 – Jun 2025*

- Designed research-matching platform and scaled to 3 continents, 20+ US states, and 2K+ users
- Secured $6.5K in grants (STARTedUP foundation) and helped scale to $38K revenue in 8 months

PROJECTS

ERCOT Climate-Risk Forecasting | *Independent (with K. Rosenfeld)* *Jan 2024 – Present*

- Modelled ERCOT outage and efficiency risk on GFDL-CM4 projections, 24 years of EIA data, and 15+ variables
- Built SARIMA-XGBoost ML pipeline (reaching 93.7% accuracy) for plant efficiency and forecasted $29.8B consumer loss scenario using Gaussian-process price models
- Published two peer-reviewed papers in *Energy Reports*; presented to MISO (Midwest grid serving ~45M people)

Boardgame Market-Success Prediction | *University of Chicago Summer Session* *July 2024 - Aug 2025*

- Analyzed 20,000 games (50+ variables) from BoardGameGeek using ensemble learning (model reached 0.71 accuracy, up from 0.39 at regression baseline) and Gini scores
- Published in Springer *LNCS* (presented at International Conference on Entertainment Computing, Tokyo 2025)

HONORS

- Top 6 Int'l Finalist, MathWorks Math Modeling (M3) Challenge | Top 0.76%, won $5K, presented at Jane Street HQ (2025)
- Top 3 Nat'l Finalist, Sempra Foundation Climate Resilience Award (2025)
- 3rd Place, Innovate WithIN | 3rd/1,500+ teams, won $4.5K in startup grants (2025)
- 2nd Place, TSA National Conference | Debating Technological Issues (2025)
- 5 peer-reviewed publications | Elsevier *Energy Reports* (2026, 2026), Springer *LNCS* (2025), *Scholarly Review* (2024, 2026)

TECHNICAL SKILLS

- Worked with Python, Java, OCaml, Typescript
- Multi-agent architecture with human-in-the-loop, context and memory engineering (Obsidian, knowledge systems), agentic coding (Claude Code, Codex, Cursor)
- AI/DS/Machine Learning (Google Colab and Python tools like scikit-learn, NumPy, Pandas)
- Sleight-of-hand card magic